

# Group Project

## Mid-presentation

**Due** 20 Mar by 23:59

**Points** 50

**Submitting** a file upload

**Available** 13 Mar at 0:00 - 20 Mar at 23:59

This assignment was locked 20 Mar at 23:59.

**Time limitation: 15 minutes presentation; 5 minutes Q&A; I will stop the presentation at 15 minutes.**

**Each group must submit a project repository link (for example, GitHub).**

**Mid-presentation Requirement:**

**Dataset Criteria:** Your dataset must meet **at least one** of the Big Data 3Vs.

- **Volume (Large-scale data):**
  - The raw dataset must be larger than **5 GB**.
  - Expectation: Must demonstrate **the use of distributed processing** (e.g., MapReduce, Spark, ect.).
- **Velocity (Fast, streaming data):**
  - Dataset generated/collected at high speed requiring **streaming or near real-time analysis**.
  - Expectation: Must **implement a real-time or simulated streaming pipeline** using Kafka (or equivalent).
- **Variety (Heterogeneous data):**
  - Dataset should involve multiple types of data that require different processing methods (e.g., text + image, etc.).
  - Expectation: Must show how different formats are **integrated, preprocessed, and analyzed**

**Dataset Selection & Motivation**

- Clearly describe the dataset.
- Justify why it is relevant and sufficiently complex for a Big Data problem (meets at least one V: Volume, Velocity, or Variety).
- Explain how the dataset connects to the project goal.

## Data Processing & Cleaning

- Show initial preprocessing steps (handling missing values, outliers, transformations, integration).
- Demonstrate use of Big Data tools (HDFS, Spark, MapReduce, Kafka, etc.) in ingestion/cleaning **if applicable**.
- Use visualizations to illustrate data characteristics and cleaning outcomes.

## Exploratory Data Analysis & Insights (incl. future Direction)

- Present meaningful visualizations tied to the project goal.
- Provide insights that highlight dataset complexity and motivate future direction (e.g., ML method, real-time data processing).
- State preliminary plan for further approach (e.g., ML, or others), based on EDA findings.

## Challenges & next steps.

- Major difficulties so far (environment installation, data size issue).
- Planned improvements for final project.

## Important rule:

- Your presentation should focus on: system pipeline design, including data storage, data distribution, and data processing workflow. Clearly explain how your system works from data input to final output.
- Briefly introduce the environment setup and system integration (e.g., Hadoop/Spark environment, tools used, and how components connect), but focus on how the system operates rather than installation details.
- Present key analytical results and insights from your processing results or visualizations. Do not include every chart or output. Select only the visuals that best explain your findings.
- Focus on system design, processing logic, and interpretation of results, rather than explaining code line by line.
- The last slide must include evidence of each member's contribution (e.g., GitHub commit history) and the **project repository link**.

project repository link.

### Mid-presentation: Grading Rubric (40 points)

| Criteria                                                                        | Excellent                                                                                                                                                                                                                                | Adequate                                                                                                               | Weak                                                                                                    | Poor                                                                                 |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Dataset Selection &amp; Motivation</b>                                       | <b>5pts:</b> Dataset clearly described, <b>strongly tied to project goal</b> , and <b>sufficiently complex</b> (large-scale, messy, streaming, or multi-modal). Well-justified choice aligned with Big Data concepts.                    | <b>3–4pts:</b> Dataset described with some justification; moderate complexity; some relevance to goal.                 | <b>1–2pts:</b> Dataset small, overly clean/simple; weak tie to Big Data concepts or project goal.       | <b>0pts:</b> Dataset trivial, irrelevant, or not explained.                          |
| <b>Data Processing &amp; Cleaning; Exploratory Data Analysis &amp; Insights</b> | <b>9–10pts:</b> Clear explanation of preprocessing steps; <b>≥4 meaningful visualizations</b> tied to project goal; insights highlight <b>dataset complexity</b> ; <b>preliminary future direction logically based on EDA findings</b> . | <b>6–8pts:</b> Basic preprocessing shown; 2–3 visuals; some insights but uneven; future direction mentioned but vague. | <b>3–5pts:</b> Minimal preprocessing; Few visuals; weak/superficial insights; future direction unclear. | <b>0–2pts:</b> No preprocessing shown. No visuals; no insights; no future direction. |

### Mid-presentation: Grading Rubric (40 points)

| Criteria                                          | Excellent                                                                                                                                                        | Adequate                                                                                                 | Weak                                                                | Poor                                                                                    |
|---------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Big Data Tool Coverage (In-Class)</b>          | <b>9–10pts:</b> <b>At least one Big Data tool covered in class</b> are clearly used. Roles of the tool in the pipeline are clearly explained and well-justified. | <b>6–8pts:</b> Big data tool mentioned but integration or justification is limited or partially unclear. | <b>3–5pts:</b> Big data tool weakly involve in the procedure        | <b>0–2pts:</b> No meaningful use of Big Data tools, or tools unrelated to course scope. |
| <b>Preliminary Modeling / Analytical Approach</b> | <b>5pts:</b> Appropriate analytical or modeling approach proposed; clearly motivated by EDA; feasible and aligned with project goal.                             | <b>3–4pts:</b> Basic approach proposed; some alignment but limited justification.                        | <b>1–2pts:</b> Approach unclear, weakly motivated, or not feasible. | <b>0pt:</b> No analytical or modeling approach proposed.                                |

### Mid-presentation: Grading Rubric (40 points)

| Criteria                                            | Excellent                                                                                                                                                                        | Adequate                                                                                   | Weak                                                                             | Poor                                                                                 |
|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
| <b>Presentation Quality &amp; Team Contribution</b> | <b>5pts:</b> Clear, professional, and engaging delivery; slides well organized; all team members contribute and present. Show project management tool.                           | <b>3–4pts:</b> Mostly clear; some delivery issues; uneven team participation.              | <b>1–2pts:</b> Weak delivery; some members silent; slides confusing.             | <b>0pt:</b> Very poor delivery; no collaboration evident.                            |
| <b>Discussion &amp; Q&amp;A</b>                     | <b>5pts:</b> Demonstrates strong understanding of dataset, data pipeline, integrate big data tool; answers questions clearly and accurately; engages thoughtfully in discussion. | <b>3–4pts:</b> Answers most questions adequately; minor gaps in explanation or confidence. | <b>1–2pts:</b> Few visuals; weak/superficial insights; future direction unclear. | <b>0pt:</b> Unable to answer questions or no meaningful participation in discussion. |